

Basic Information

Experiment No.: 07

Date: 24-09-2026

Title: Java Hashtable and Enumeration

Objective / Aim

To implement "Hashtable" in Java and demonstrate key-value operations, searching, removal, size checking, and traversal using "Enumeration".

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

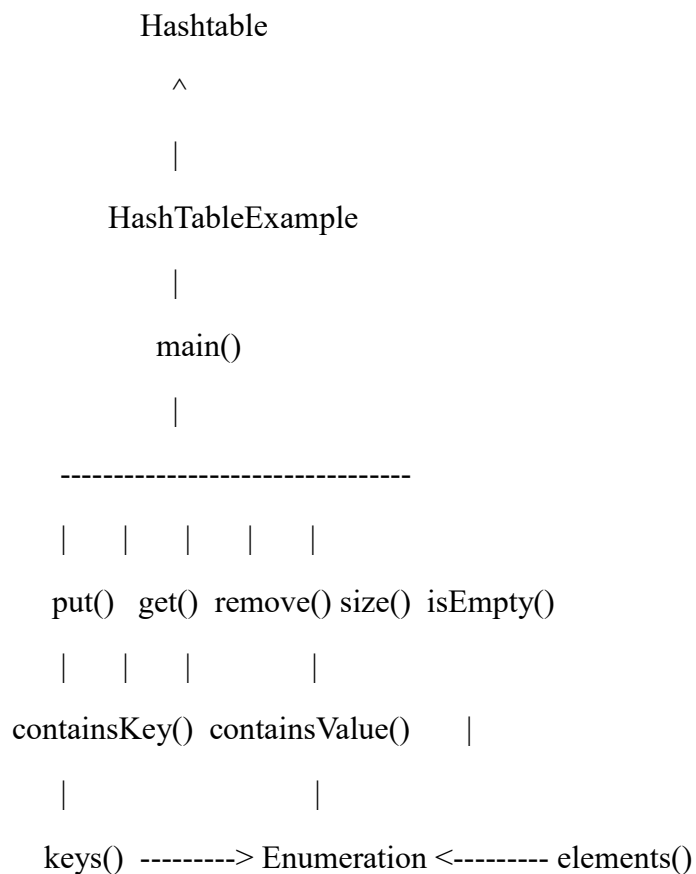
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "Hashtable" with integer keys and string values.
2. Add key-value pairs using "put()".
3. Access a value using "get()".
4. Check keys and values using "containsKey()" and "containsValue()".
5. Remove an entry using "remove()".

6. Check the size and empty status of the table.
 7. Get keys using "keys()".
 8. Traverse keys using "Enumeration".
 9. Get and traverse values using "elements()" and "Enumeration".
 10. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.Hashtable;
import java.util.Enumeraation;

public class HashtableExample {
    public static void main(String[] args) {

        Hashtable<Integer, String> table = new Hashtable<>();

        // put()
        table.put(101, "Rahul");
        table.put(102, "Priya");
        table.put(103, "John");

        System.out.println("Hashtable: " + table);

        // get()
        System.out.println("Value of key 102: " +
            table.get(102));

        // containsKey()
        System.out.println("Contains key 101: " +
            table.containsKey(101));

        // containsValue()
        System.out.println("Contains value John: " +
            table.containsValue("John"));

        // remove()
```

```
table.remove(103);

System.out.println("After remove: " + table);

// size()
System.out.println("Size: " + table.size());

// isEmpty()
System.out.println("Is Empty: " + table.isEmpty());

// keys()
System.out.println("\nKeys:");

Enumeration<Integer> keys = table.keys();

while (keys.hasMoreElements()) {
    System.out.println(keys.nextElement());
}

// elements()
System.out.println("\nValues:");

Enumeration<String> values = table.elements();

while (values.hasMoreElements()) {
    System.out.println(values.nextElement());
}
}
```

Compilation & Execution Commands

Save the file as:

HashTableExample.java

Since the program contains:

```
package collectionframework;
```

the file should be inside the "collectionframework" folder.

Compile:

```
javac HashTableExample.java
```

Run:

```
java collectionframework.HashTableExample
```

Output Screen

Note: "Hashtable" does not guarantee a particular display/traversal order, so the order of entries, keys, and values may vary.

Hashtable: {103=John, 102=Priya, 101=Rahul}

Value of key 102: Priya

Contains key 101: true

Contains value John: true

After remove: {102=Priya, 101=Rahul}

Size: 2

Is Empty: false

Keys:

102

Values:

Priya

Rahul

Result

The "Hashtable" class was successfully implemented, and operations such as "put()", "get()", "containsKey()", "containsValue()", "remove()", "size()", and "isEmpty()" were demonstrated. Keys and values were successfully traversed using "Enumeration".